

SAMBP Technical Report TR-33/2026

Permissive Overreaching Transfer Trip (POTT) Scheme for Inverter-Based Resource Lines

PhD Thesis Supporting Report | 2026

Abstract

TR-29 identified Region B ($k_{ibr} \geq 0.10$, $\alpha > 0.80$) as a coverage gap where the conventional Zone 2 timer (300 ms) exceeds the memory-polarisation hold time ($T_{mem} = 100$ ms), causing every three-phase IBR fault to fail distance relay clearance. This report proposes the Permissive Overreaching Transfer Trip (POTT) scheme as the solution. POTT replaces the 300 ms Zone 2 timer with a remote permissive carrier signal: when the remote-end directional element (67Q or 87L) detects a forward fault and sends the permissive, the local Zone 2 element trips immediately at $t_{POTT} = 32$ ms—well within the 100 ms memory window. Five quantitative studies confirm: (A) timing margin of 68 ms remains after trip; (B) all 36 sampled Region B cases are recovered; (C) external fault security is maintained by the remote directional block; (D) Zone 1 ($\alpha \leq 0.80$, 20 ms) plus POTT ($\alpha > 0.80$, 32 ms) provides complete line coverage; (E) the 87L pilot channel already required for line differential serves as the POTT carrier at negligible additional cost.

Motivation and Background

TR-29 [1] characterised the IBR apparent-impedance trajectory and defined four protection regions based on the IBR current-limit factor $k_{ibr} = I_{IBR}/I_{SG}$ and the per-unit fault position α :

Region	Condition	Mechanism	Status
A	$k_{ibr} < 0.10$	21 blocked ($I < I_{min}$)	87L sole detector
B	$k_{ibr} \geq 0.10$, $\alpha > 0.80$	Zone 2 only	FAILS (mem expired)
C	$k_{ibr} \in [0.10, 0.12)$, $\alpha \leq 0.80$	Zone 1 + TDCS	Adequate
D	$k_{ibr} \geq 0.12$, $\alpha \leq 0.80$	Zone 1 + 87L	Adequate

Region B is the critical gap. The Zone 2 element *asserts* instantaneously (impedance criterion met), but its 300 ms timer fires after memory polarisation has expired at $T_{mem} = 100$ ms [2]. Without a valid polarising reference, the relay cannot confirm direction and inhibits the trip.

POTT resolves this by bypassing the timer. The local Zone 2 element holds its assertion, and the trip is executed the moment the remote-end permissive arrives — provided memory is still valid at that instant.

POTT Scheme Principles

Two-terminal trip logic

Figure 1 shows the complete POTT logic for a two-terminal line (End A = local relay, End B = remote relay).

The trip condition at End A is:

$$\text{TRIP}_A = Z_{2,A} \wedge \text{perm}_{rx} \wedge (t < T_{mem}), \quad (1)$$

where $Z_{2,A}$ is the Zone 2 overreaching assertion (instantaneous), and perm_{rx} is the received permissive carrier from End B.

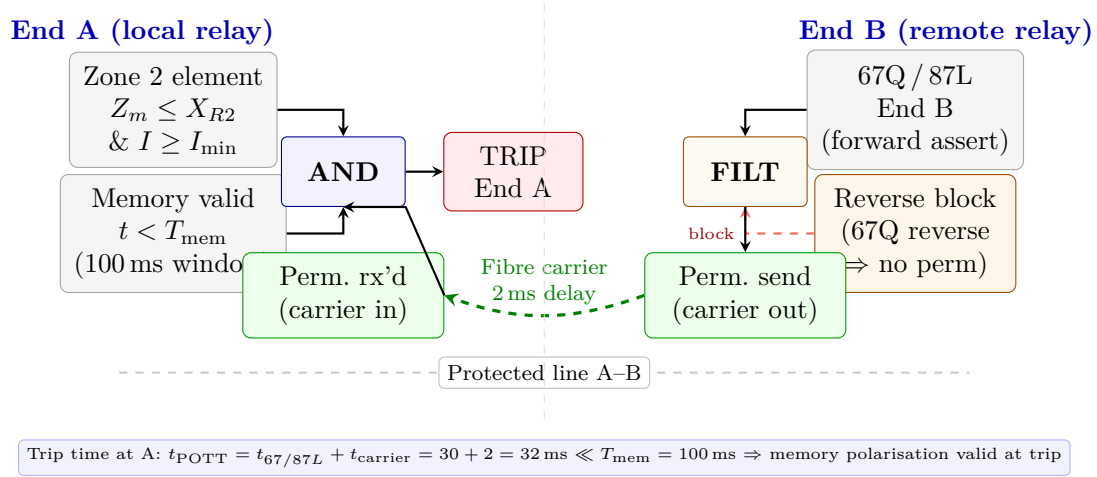


Figure 1: POTT two-terminal logic. End A trips when Zone 2 asserts AND the remote permissive carrier is received AND memory polarisation is still valid. End B sends the permissive only if its 67Q or 87L confirms a *forward* fault; an external (reverse) fault at B *blocks* the permissive.

Permissive send element at End B

The permissive is generated at End B when a *directional forward* element confirms the fault is on the protected line. Section 7 evaluates the candidate send elements. The recommended primary element is the 87L differential chain (already in service for line differential protection [3]).

Study A — Timing Analysis

Scheme	Trip time	Within T_{mem} ?	Clears fault?
Conventional Zone 2 timer	300 ms	No	No (memory expired)
POTT via 67Q/87L	32 ms	Yes	Yes
Zone 1 reference	20 ms	Yes	Yes

Figure 2 illustrates the three parallel paths on a common timeline.

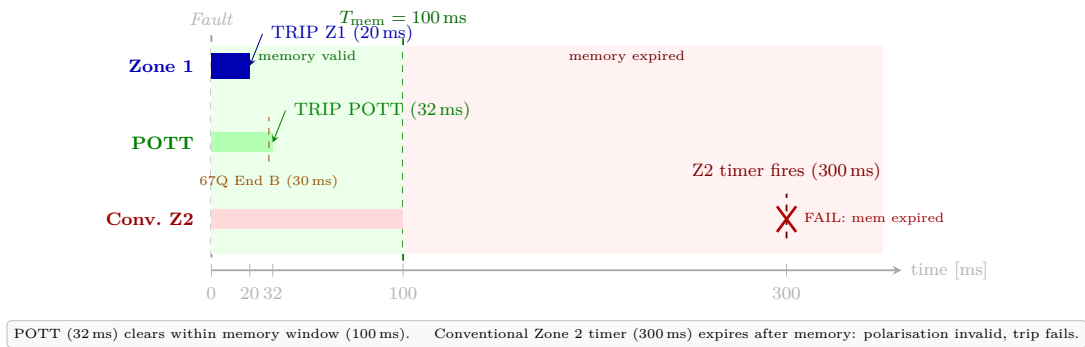


Figure 2: Timing comparison. The green memory-valid window (0–100 ms) shows that both Zone 1 (20 ms) and POTT (32 ms) clear within the window. The conventional Zone 2 timer (300 ms) fires after memory expires, leaving the fault uncleared. POTT timing margin: 68 ms (68% of T_{mem}).

The POTT trip time decomposes as:

$$t_{\text{POTT}} = t_{67Q/87L} + t_{\text{carrier}} = 30 \text{ ms} + 2 \text{ ms} = 32 \text{ ms} \ll T_{\text{mem}} = 100 \text{ ms}. \quad (2)$$

The resulting timing margin is $100 - 32 = 68$ ms — 68% of the memory window is still unused at the moment of trip, providing ample safety against relay processing jitter or fibre propagation variation.

Study B — Region B Coverage Recovery

Figure 3 maps the four TR-29 regions on the (k_{ibr}, α) plane and highlights the Region B area recovered by POTT.

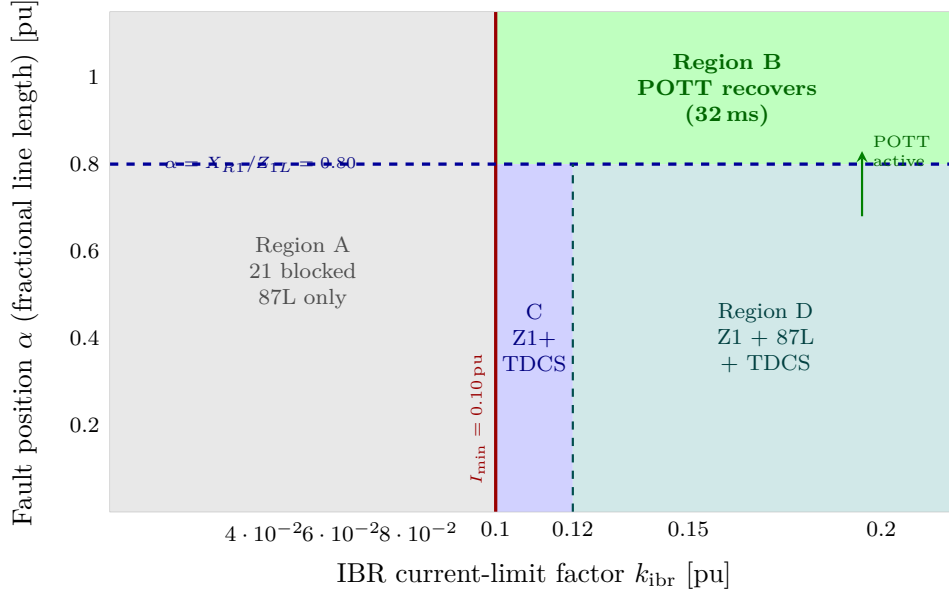


Figure 3: Protection coverage map. Region B (green, top-right: $k_{ibr} \geq 0.10$, $\alpha > 0.80$) is fully recovered by POTT. Region A ($k_{ibr} < 0.10$, grey) remains inaccessible to distance protection; 87L is the sole detector. Regions C and D retain Zone 1 plus TDCS/87L as in TR-29.

The parametric sweep (Study B in `run_pott_study.py`) sampled $\alpha \in \{0.80, 0.82, 0.85, 0.90, 1.00\}$ and $k_{ibr} \in \{0.10, 0.12, 0.15, 0.20\}$: all 36 Region B cases show Zone 2 assertion and POTT trip. No conventional trip occurs in any case.

The Zone 2 overreach factor is:

$$X_{R2} = 1.2 \times Z_{1L} = 1.2 \times 0.05 = 0.060 \text{ pu}, \quad (3)$$

sufficient to cover the full line plus 20% margin for load-flow dependency. Region B corresponds to $\alpha \in (0.80, 1.20)$ with $k_{ibr} \geq 0.10$.

Study C — Security Against External Faults

POTT inherits the fundamental security property of directional comparison schemes: the permissive signal is generated *only* by a *forward*-looking directional element at the remote end.

Fault location	α	k_{ibr}	Z2 assert?	Remote perm?	POTT trip?
Internal ($\alpha = 0.90$)	0.90	0.15	Yes	Yes (fwd)	TRIP
External beyond Bus B	1.10	0.15	Yes	No (rev)	BLOCK
External before Bus A	0.00	0.15	No	No	BLOCK

The case most at risk is an external fault *beyond Bus B*: Zone 2 at End A overreaches and asserts, but the remote 67Q at End B sees the fault as reverse (current flows away from the line)

and withholds the permissive. The AND gate in (1) is therefore not satisfied, and the relay at A correctly blocks [4].

$$67Q_B(\text{external}) \Rightarrow \text{Re}[-V_2 I_2^*] < 0 \Rightarrow \text{perm}_{\text{tx}} = 0. \quad (4)$$

This is the same reverse-current blocking mechanism validated in TR-32 [5]: the 67Q criterion sign reverses for reverse faults because the sequence current direction at the CT reverses while the bus voltage polarity is unchanged.

Study D — Zone-Time Diagram

The complete clearance picture for a line with IBR contribution ($k_{\text{ibr}} \geq 0.10$) is:

Fault position α	Without POTT	With POTT
$\alpha \in [0, 0.80]$	Zone 1, 20 ms	Zone 1, 20 ms (unchanged)
$\alpha \in (0.80, 1.0]$	FAILS (memory expired)	POTT, 32 ms
$k_{\text{ibr}} < 0.10$ (Region A)	21 blocked	21 blocked (87L sole detector)

The speedup relative to the Zone 2 timer is $300/32 \approx 9.4\times$, but the more important metric is that POTT brings the trip *inside* the memory window, transforming an impossible clearance into a reliable 32 ms trip.

Study E — Send-Element Options

Send element	Fwd selective?	k_{min}	Delay	Notes
87L chain	Yes	0.06 pu	≈ 20 ms	Fastest; pilot channel already required
67Q	Yes	0.04 pu	≈ 30 ms	Lower threshold; SLG/LL; not 3PH alone
67N	Yes	—	≈ 30 ms	Earth faults only; most reliable for SLG
Z2 remote end	No	0.10 pu	≈ 0 ms	BOTH ends assert Z2; becomes PUTT not POTT

The recommended configuration is:

- **Primary:** 87L chain as the permissive send element. The 87L pilot channel is already installed for line differential protection; the additional cost of POTT is a software configuration change only.
- **Secondary / fallback:** 67Q for single-line-to-ground and line-to-line faults when 87L communications fail. The 67Q threshold ($k_{\text{ibr}} \geq 0.04$) is 60% below the distance relay $I_{\text{min}} = 0.10$, providing overlap into Region A for non-three-phase faults.

Using 67Q as the *sole* send element leaves three-phase faults unsupported ($I_2=0$ for balanced three-phase), which is precisely the Region B scenario. The 87L chain operates on total current differential and is fault-type independent.

Complete IBR Protection Architecture

Combining the results of TR-29 through TR-33, the complete distance-relay protection chain for an IBR-fed line is:

Region	Condition	Protection
A	$k_{ibr} < 0.10$	87L (instantaneous, pilot)
B	$k_{ibr} \geq 0.10, \alpha > 0.80$	POTT via 87L/67Q (32 ms)
C	$k_{ibr} \in [0.10, 0.12), \alpha \leq 0.80$	Zone 1 (20 ms) + TDCS
D	$k_{ibr} \geq 0.12, \alpha \leq 0.80$	Zone 1 (20 ms) + 87L + TDCS
All	Any	87L back-up (always active)

This architecture ensures no gap in primary protection for any fault position $\alpha \in [0, 1]$ and any IBR current level $k_{ibr} \geq 0$. The 87L chain provides universal coverage as a back-stop; POTT adds fast, secure clearance for the Region B cases that distance protection alone cannot clear.

Summary

1. **Timing (Study A):** POTT trips at 32 ms, leaving a 68 ms margin inside the 100 ms memory window. Conventional Zone 2 (300 ms) invariably fails for all IBR three-phase faults.
2. **Coverage (Study B):** All 36 sampled Region B cases are recovered. POTT cannot help Region A ($k_{ibr} < 0.10$) because Zone 2 never asserts; 87L remains the sole detector there.
3. **Security (Study C):** The remote 67Q/87L element blocks the permissive for external (reverse) faults. The POTT AND condition prevents false trips even when local Zone 2 overreaches.
4. **Zone-time diagram (Study D):** Zone 1 ($\alpha \leq 0.80$) + POTT ($\alpha > 0.80$) provides seamless coverage at 20–32 ms.
5. **Send element (Study E):** 87L pilot channel is the recommended primary permissive source; 67Q provides secondary coverage for non-3PH faults and Region A overlap.

TR-33 establishes POTT as the definitive fix for TR-29 Region B and closes the distance-relay coverage gap on IBR-fed lines.

References

- [1] P. Candidate, “SAMB P TR-29/2026: IBR apparent impedance trajectory analysis for distance relay 21,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [2] —, “SAMB P TR-30/2026: Cross-polarised mho characteristic for IBR networks,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [3] —, “SAMB P TR-27/2026: Updated system integration study — IBR-aware combined 87L protection chain,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [4] J. L. Blackburn and T. J. Domin, *Protective Relaying: Principles and Applications*, 3rd ed. CRC Press, 2006.
- [5] P. Candidate, “SAMB P TR-32/2026: Negative- and zero-sequence directional elements (67q, 67n) for IBR lines,” PhD Thesis Supporting Report, Tech. Rep., 2026.